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Attention Under Digital Distraction: An Eye-Tracking Study on the Effectiveness of Blocking Interventions

Short Paper

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Abstract

Individuals are increasingly exposed to digital distractions with potential negative consequences. While blocking interventions can be mitigators, little is known about how different blocking modes and distractors influence attentional mechanisms and task performance. Drawing on the process model of self-control, we examine how self-deployed and other-deployed blocking affect attentional mechanisms and task outcomes under digital distraction. We conducted a laboratory experiment with eye-tracking to capture real-time attentional dynamics during a reading comprehension task. Participants were assigned to four conditions: no distraction, distraction without intervention, distraction with self-deployed blocking, and distraction with other-deployed blocking. Results show that digital distractions reduce task attentional engagement, while blocking interventions reduce attentional engagement with distractors. Self-deployed blocking leads to lower attentional capture and dwelling on distractions compared to other-deployed conditions. The findings advance our understanding of digital distractions by revealing underlying attentional mechanisms and highlighting the importance of user autonomy in designing attention management interventions.

Keywords: Digital Distraction, Attention, Eye-Tracking, Blocking Interventions, Self-control

Introduction

As work and learning increasingly take place in digital environments, individuals are persistently exposed to digital distractions such as advertisements and other task-irrelevant stimuli. These distractions compete for attention and often divert focus away from the task at hand (Taylor et al. 2024). While brief diversions may occasionally provide relief from monotonous work (Puranik et al. 2020), sustained exposure to digital distractions can impair task performance, prolong task completion time, and negatively affect well-being (Ophir et al. 2009; Rosen & Samuel 2015). In online task settings, where distractions are typically visually salient and continuously embedded into people's screens, managing attention becomes particularly challenging, raising practical concerns for individuals, educators, and organizations alike.

In this area, prior information systems research has largely examined digital disruptions in the form of IT interruptions, such as emails or instant messages (Addas & Pinsonneault 2015, 2018). However, interruptions differ from digital distractions. *Interruptions* involve a purposeful shift of attention to a

secondary task, whereas *distractions* involve involuntary shift of attention to task-irrelevant stimuli that ideally should be ignored (Clapp & Gazzaley 2012). Prior work, while not always distinguishing between the terms clearly, has considered digital distraction as a self-control problem, where individuals must resist immediately rewarding stimuli in favor of longer-term task goals (Duckworth et al. 2016; Inzlicht et al. 2021). Process models of self-control suggest that interventions targeting early stages of this process, particularly by modifying the situation to remove tempting stimuli, are more effective than those relying on late-stage impulse suppression.

To mitigate the effects of digital distractions, several intervention tools have been proposed including blocking, restriction, and feature modification (Biedermann et al. 2021; Roffarello & De Russis 2023). Among these, *blocking interventions*, those that prevent access to distracting stimuli altogether, are especially relevant because they operate at the early, situation stage of the self-control process. Importantly, such interventions can be deployed in two main ways. They may be *self-deployed*, where individuals voluntarily choose to block distractions, or *other-deployed*, where blocking is imposed externally, for example by systems or organizational administrators (Duckworth et al. 2018). These two models of deployment differ fundamentally in the degree of autonomy and control afforded to users and are therefore likely to have differential effects on attention.

Despite the potential of blocking interventions, empirical evidence comparing the effectiveness of self-deployed versus other-deployed blocking remains limited (Biedermann et al. 2021; Lyngs et al. 2022). Moreover, in terms of distraction outcomes, existing studies have largely relied on self-reported attention or post-task assessments (Roffarello & De Russis, 2023), which are susceptible to recall bias and offer limited insights into how attention dynamically shifts during task execution. Eye tracking technologies offer a methodological refinement by enabling direct, objective measurement of visual attention through metrics such as fixations, dwell time, saccades, blink rate, and pupil dilation, measures that have been shown to reliably capture attentional allocation and cognitive load (Vasseur et al. 2023).

Motivated by these important gaps, this study examines the following three *research questions*: (1) How can eye-tracking assess key attentional mechanisms during digital distraction? (2) Does digital distraction affect individual attentional mechanisms? and (3) Are self-deployed blocking interventions more effective than other-deployed blocking interventions in countering the effects of digital distractions on attentional mechanisms? To address these questions, we build on a process model of digital distraction and self-control and conduct a laboratory experiment. We focus on digital distractions in our study and examine the impact of visually distracting stimuli on attentional mechanisms. By employing eye-tracking-based measures of attention within an experiment, this research offers salient contributions by providing a process-oriented understanding of how digital distractions and different modes of blocking intervention deployment shape attention and performance.

Conceptual Background

Digital Distractions and Interventions

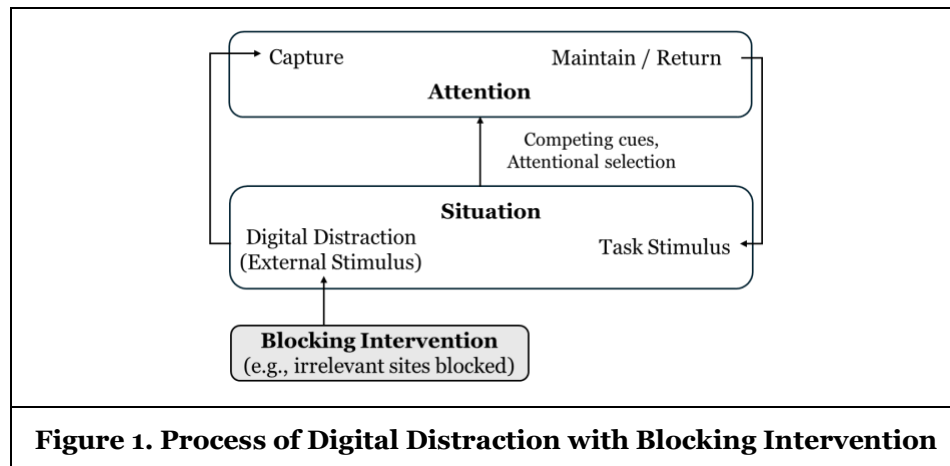
Digital distractions are digital interferences that are irrelevant to the task at hand yet capable of drawing individuals' attention away from their focal task (Clapp & Gazzaley 2012). Digital technologies are intentionally designed to capture and sustain attention through features such as notifications and advertisements, increasing engagement while also heightening susceptibility to distraction (Eyal 2013; Bhargava & Velasquez 2021; Lee et al. 2025). Although prior research has examined related phenomena such as multitasking and IT interruptions (Addas & Pinsonneault 2015, 2018), digital distraction remains under-theorized in the information systems literature, and empirical evidence on the effectiveness of mitigation strategies is limited (Hanin 2021; Roffarello & De Russis 2023).

To explain how digital distractions arise and how they may be mitigated, we draw on the concept of self-control. *Self-control* refers to the capacity to regulate one's thoughts and behaviors in the face of conflicts between immediately rewarding impulses and longer-term goals (Duckworth et al. 2016; Inzlicht et al. 2021). In technology-rich environments, digital distractions heighten such conflicts by presenting salient, tempting stimuli that compete with task objectives, making lapses of self-control more likely (Hanin 2021). We adopt a *process model of self-control* (Duckworth et al. 2016; Inzlicht et al. 2021), which conceptualizes control and regulation as unfolding in stages over time—from the emergence of a temptation to the

allocation of attention and the behavioral response to it. This perspective allows us to theorize both how attentional diversion arises from digital distractors and how interventions, such as blocking, can modify the situation before distraction occurs. Thus, to explain how digital distractions influence attention as well as the effect of blocking interventions, we build on the process model of self-control (Duckworth et al. 2016; Inzlicht et al. 2021).

Process Model for Digital Distraction

The process model of self-control posits that impulses develop in an iterative cycle and illustrates the stages through which they arise and are either augmented or weakened over time (Duckworth et al. 2016). Drawing on the self-control process model we focus on the situation and attention stages where digital distractions are first experienced and then influence attentional mechanisms followed by task outcomes. This is because these are the first two stages that show the emergence of digital distractions as well as the effects of blocking interventions in task environments, thereby forming the initial critical pathway for involuntary attentional diversion (see Figure 1).



The *situation* stage is when external, task-irrelevant digital stimuli (e.g., pop-up advertisements, social media notifications) appear in users' task interface. This is followed by the immediate *attention* stage where these typically visually salient stimuli trigger reflexive attentional capture, pulling users' focus away from their task and disrupting sustained attentional engagement. This unmediated situational-attentional dynamic creates the foundational condition for digital distraction, with the presence of distractors in the situation stage directly shaping attentional allocation at the attention stage.

Aligning with self-control theory's emphasis on early situational modification as the most effective form of self-regulation (Duckworth et al. 2016), blocking interventions target the situation stage exclusively, intervening before distractors can elicit attentional capture at the attention stage. By removing or restricting access to digital distractors in the task environment, these interventions disrupt the link between the situational presence of distractors and subsequent attentional diversion. We further distinguish between two deployment modes of such situation-stage blocking interventions. These are self-deployed blocking, where users voluntarily initiate and control the removal of distractors, and other-deployed blocking, where distractor removal is externally imposed by systems or administrators (Duckworth et al. 2018). These two modes differ fundamentally in the degree of user autonomy afforded at the situation stage, where autonomy variation is posited to shape subsequent attentional dynamics at the attention stage—including the strength of attentional capture and the ability to maintain focal attention on the task—thereby driving divergent effects on attentional regulation in digital settings.

Attentional Mechanisms and Eye Tracking

Having conceptualized digital distraction as a process unfolding across situation and attention stages leading to subsequent appraisal and responses, a central methodological challenge concerns how to capture attentional mechanisms as they occur. Drawing from attention research, three mechanisms are central: *attentional capture*, *attentional engagement*, and *attentional dwelling*. *Attentional capture* refers to the

rapid, initial orienting of visual attention toward a stimulus (Remington et al., 2001). In digital environments, visually salient elements such as advertisements are designed to trigger such automatic orienting responses. Following capture, attention may transition into *attentional engagement*, defined as the resource-demanding allocation of cognitive processing to a selected stimulus (Zivony et al., 2018). Finally, *attentional dwelling* refers to the sustained allocation of attention at a selected location, reflecting delayed disengagement from a stimulus (Lamy et al., 2018). Together, these mechanisms conceptualize distraction as a dynamic sequence of orienting, processing, and sustained diversion from the focal task.

Because these mechanisms unfold moment-to-moment, they are not fully observable through retrospective self-reports of concentration. Eye tracking provides a well-established and objective method for measuring visual attention as it unfolds in real time. Prior research demonstrates that eye movement indicators provide reliable proxies of attentional allocation and cognitive load (Buettner et al., 2018; Vasseur et al., 2023), and fixation patterns reveal which on-screen elements capture and sustain attention (Bera et al., 2019; George et al., 2022). By capturing gaze allocation and dwell time (George et al. 2022), eye tracking enables fine-grained analysis of attentional diversion that cannot be obtained through behavioral outcomes or self-report alone. Accordingly, eye tracking offers us a methodologically aligned approach for examining the attentional processes underlying digital distraction. The above conceptual background supports our hypotheses framing.

Hypotheses

Digital distractions are designed to capture visual attention and are therefore expected to divert users' gaze away from their focal task. When distractors are present, users are expected to shift their gaze away from the focal task toward the distractors. Such attentional diversion consumes cognitive resources and creates switching costs (Altmann et al., 2014; Orhan et al., 2021). Cognitive switching disrupts task flow, increases processing demands, and reduces efficiency. Accordingly, we propose that digital distractions should impair attentional engagement with the task.

H1: *Users will exhibit greater attentional engagement to the task when there is no distractor as compared to when there are distractors.*

Blocking interventions operate at the situation stage of the self-control process by removing distracting stimuli before attentional capture occurs (Duckworth et al. 2016). By limiting exposure to visually salient distractors, blocking interventions are expected to reduce attentional engagement with distracting content. Without blocking, users are expected to exhibit sustained and repeated visual engagement with distracting stimuli. Such sustained attentional diversion increases cognitive switching costs and disrupts task flow (Altmann et al., 2014; Orhan et al., 2021). Thus, we propose that,

H2: *Users will exhibit greater attentional engagement to distractors when there is no blocking intervention as compared to when there are blocking interventions.*

Although both self-deployed and other-deployed blocking interventions remove distracting stimuli, they differ in the degree of autonomy and control afforded to users (Biedermann et al. 2021; Lyngs et al. 2022). Self-control theory suggests that self-deployed interventions are less resource-depleting and more effective at regulating attention than externally imposed interventions (Muraven 2008). Rigid blocking and denying access to digital distractions to users may lead to resource depletion, resulting in worse self-control (Baumeister et al. 1994, 2018). Prior work further suggests that attentional disengagement reflects users' ability to maintain focus in the presence of potential distractors. Thus, for self-deployed blocking interventions, the attentional capture by and dwelling on distractors are expected to be weaker. Accordingly, better task performance is hypothesized for self-deployed blocking interventions.

H3a: *Self-deployed blocking interventions will be associated with lower attentional capture by distractors as compared to other-deployed blocking interventions.*

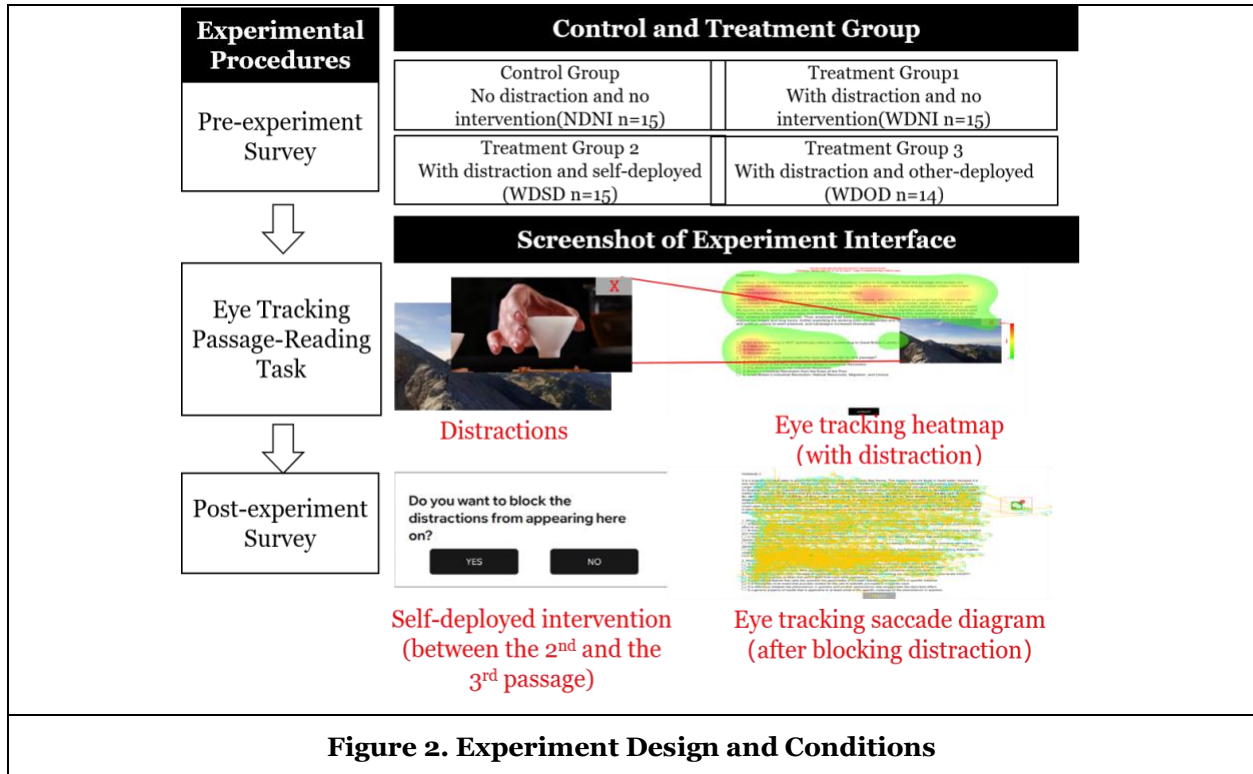
H3b: *Self-deployed blocking interventions will be associated with lower attentional dwelling on the distractor as compared to other-deployed blocking interventions.*

H3c: *Self-deployed blocking interventions will be associated with shorter task completion time and higher task performance score as compared to other-deployed blocking interventions*

Experiment Design and Procedures

Experiment Design

We adopted a between-subjects experimental design to test the hypotheses. Four distinct experimental conditions were established to systematically compare the effects of digital distractions and blocking interventions (see Figure 2). The conditions consisted of the control group with no-distraction-no-intervention (NDNI), the treatment group with-distraction-no-intervention (WDNI), as well as the treatment groups with-distraction-self-deployed (WDSD) blocking or with-distraction-other-deployed (WDOD) blocking.



As our subjects were students from a Chinese University, we chose a task relevant for students in general. The experimental task was an English-language online reading comprehension task adapted from standardized test assessments. As common in these assessments, the task included 4 passages (800–1000 characters each) and 11 multiple-choice questions that evaluated literal understanding, inferential reasoning, and information synthesis.

The digital distractions consisted of visually salient, task-irrelevant pop-up stimuli (e.g., promotional ads) displayed in the upper-right corner of the screen. These distractors appeared at fixed intervals—30s, 90s, 150s, and 210s after the start of each passage—and remained visible until manually dismissed by the participant within a few seconds. The blocking intervention, which transformed salient advertisements into inconspicuous icon-form notifications, was introduced between the second and third reading passages. While the transformation itself was identical across groups, the deployment method varied to test the effects of user autonomy. Participants in the self-deployed blocking group (WDSD) were given the discretion to choose and initiate the intervention, allowing them to maintain full control over its activation. In contrast, the intervention for the other-deployed blocking group (WDOD) was externally imposed by the system, which automatically activated the blocking mechanism for all subsequent passages without any user control. Following prior eye-tracking research in information systems and human-computer interaction that undertook controlled laboratory studies (e.g., Bera et al. 2019), we recruited approximately 15 participants per condition to examine attentional processes and gaze behavior.

The experiment setup was implemented via Psycopy software to precisely control distraction stimulus presentation, timing, and data recording. The experimental workflow lasted approximately 50 minutes, following a standardized protocol. Participants first completed an informed consent form and a pre-survey for measuring control variables (self-control, multitasking tendency, stress level), followed by eye tracking calibration to ensure gaze accuracy. They then completed the four reading passages sequentially, with continuous recording of eye tracking data and performance data (task completion time, comprehension scores). After the task, participants finished a post-task survey assessing perceived distraction, intervention usability, and task difficulty, following which they were debriefed and compensated with 50 Chinese Yuan.

Eye Tracking Setup and Measures

To capture visual attention dynamics during the task, we employed a EYELINK portable Duo eye-tracking system (120 Hz), a high-precision instrument widely utilized for non-invasive measurement in human-computer interaction and cognitive science (Vasseur et al. 2023). A height-adjustable chair and a headrest were used to minimize head movement and postural variability and ensure data accuracy. Prior to the task, a 9-point calibration was conducted to align the system with each participant's eye position. The eye-tracking system was synchronized with the Psycopy software to integrate gaze data with task events (e.g., distractor appearance). Eye-tracking data were analyzed within a 5-second post-distractor window, a timeframe that captures the critical phase of initial attentional capture (Clapp & Gazzaley 2012). The operationalizations of the core attentional constructs are detailed in Table 1.

Table 1. Constructs with Eye-tracking Indicators			
Construct	Indicators	Operationalization	Theoretical Rationale
Attentional Capture	First Fixation Duration (ms)	Duration of the first stable gaze on distractors after their appearance	Longer first fixation indicates stronger attentional capture (Buettnier et al. 2018)
Attentional Engagement	Dwell Time (ms)	Total duration of fixations directed at distractors within the recording window	Longer dwell time indicates that the viewed object is engaging or creates interest for the viewer (Poole & Ball 2006)
	Fixation Count	Number of discrete fixations directed at distractors	High number of fixations on a particular area indicates it is more important to the viewer than other areas (Bera et al. 2019)
Attentional Dwelling	Regression Count	Number of gaze shifts from the focal task area back to distractors	Higher regression count reflects difficulty maintaining task focus and disengagement from distractors (Vasseur et al. 2023)

Analysis and Results

Our analysis is based on pre- and post-task survey questionnaires, and eye-tracking metrics during the task. To rigorously isolate the effects of the experimental conditions and address methodological concerns regarding statistical precision in smaller samples (Miller & Chapman, 2001), we employed Analysis of Covariance (ANCOVA) for all hypothesis testing. All models controlled for trait self-control, multitasking preference, and baseline stress level, which prior research has shown to influence susceptibility to digital distraction (Duckworth et al., 2016).

We first tested H1, which predicts greater attentional engagement with the focal task in the no-distraction condition relative to distractor-present conditions. The ANCOVA revealed a significant omnibus main effect of experimental condition on task-relevant fixation count ($F(1, 52) = 12.24, p < 0.001, \eta_p^2 = 0.191$), as well as a significant main effect on task-relevant dwell time ($F(1, 52) = 7.86, p = 0.007, \eta_p^2 = 0.131$). Descriptive statistics aligned with these effects: participants in the no-distraction (NDNI) group showed substantially higher dwell time ($M = 4124.80$ vs. 3404.38 ms) and fixation counts ($M = 16.20$ vs. 9.15) on task-relevant content relative to the three distracted groups combined. Planned contrasts further confirmed these differences were statistically significant, providing robust support for H1.

We next tested H2, H3a, and H3b by restricting our ANCOVA to the three distractor-present conditions (WDNI, WDSO, WDOO, $n=44$), focusing on the data collected after the blocking intervention was activated. The omnibus ANCOVA revealed a significant main effect of condition on distractor-related dwell time ($F(1, 38) = 23.15, p < 0.001, \eta_p^2 = 0.378$) and on distractor-related fixation count ($F(1, 38) = 24.02, p < 0.001, \eta_p^2 = 0.387$). Planned contrasts further confirmed that the unmitigated distraction (WDNI) group showed substantially higher attentional engagement with distractors relative to both blocking intervention groups combined, supporting H2.

Pairwise comparisons further confirmed that the self-deployed blocking (WDSO) group had significantly weaker attentional capture i.e., shorter first fixation duration time ($F(1, 38) = 5.27, p = 0.027, \eta_p^2 = 0.122$) and reduced attentional dwelling i.e., fewer regressions back to distractors ($F(1, 38), p = 0.033, \eta_p^2 = 0.114$) relative to the other-deployed (WDOO) group (all $p < 0.05$), fully supporting H3a and H3b.

Table 2. ANCOVA Results for Eye-Tracking Metrics

Hypothesis	Dependent Variable	Analysis Sample	ANCOVA Test Statistic	Expect p value	Partial η^2	Hypothesis
H1	Task-relevant Fixation Count	NDNI vs WDNI+WDSO+WDOO	$F(1,52) = 12.24$	<0.001	0.191	Fully supported
	Task-relevant Dwell Time	NDNI vs WDNI+WDSO+WDOO	$F(1,52) = 7.86$	0.007	0.131	
H2	Distractor-related Dwell Time	WDNI vs WDSO+WDOO	$F(1,38) = 23.15$	<0.001	0.378	Fully supported
	Distractor-related Fixation Count	WDNI vs WDSO+WDOO	$F(1,38) = 24.02$	<0.001	0.387	
H3a	First Fixation Duration on Distractors	WDSO vs WDOO	$F(1,38) = 5.27$	0.027	0.122	Fully Supported
H3b	Regression Count to Distractors	WDSO vs WDOO	$F(1,38) = 4.89$	0.033	0.114	Fully Supported

Note: All ANCOVA models controlled for trait self-control, multitasking preference and baseline stress level

Regarding task performance (H3c), subjective responses indicated higher perceived concentration and performance under self-deployed blocking as compared to other-deployed blocking (Concentration: $\beta = 0.44, p = 0.042$; Performance: $\beta = 0.48, p = 0.033$). However, objective data showed no significant difference between the WDSO group and WDOO group in terms of task performance score ($p = 0.343$) and task completion time ($p = 0.701$). In other words, H3c was partially supported i.e., for perceived performance but not for objective performance. The findings suggest that the perceived effectiveness of distraction management interventions could be dependent on whether distractions are blocked, and who initiates the blocking process. Self-deployed interventions may foster a stronger sense of agency and intentional self-regulation, thereby improving subjective task experience (Duckworth et al., 2016). On the other hand, objective performance could depend on the task fit with the subjects. In our study, the lack of advanced English familiarity of the subjects may have limited their task performance irrespective of whether they were in the self-deployed blocking versus the other-deployed blocking intervention condition.

Discussion and Contributions

This study offers three key contributions to the information systems literature on digital distraction. *First*, it unveils the differential effectiveness and underlying mechanisms of self-deployed versus other-deployed blocking interventions, a critical comparison under-explored in prior studies (e.g., Biedermann et al. 2021; Roffarello & De Russis 2023). The findings support the proposition that autonomous interventions are less cognitively resource-depleting (Muraven 2008) and demonstrate that user autonomy in intervention deployment shapes core attentional regulation processes. *Second*, we establish a direct empirical link between objective eye-tracking metrics and key attentional constructs (attentional engagement, attentional capture, attentional dwelling), where operationalizing these constructs provides a methodological foundation for future attention research. *Third*, this study discovers the attentional mechanisms underlying the digital distraction process model. Digital distractions exert influence through involuntary attentional capture and impaired attentional dwelling, while situational blocking interventions mitigate distraction by intercepting this process at the early situation stage, thus strengthening the causal inference between intervention design and self-control outcomes (Markus & Robey 1988). Practically, our findings inform the design of user-centric tools for digital work and learning environments, highlighting that voluntary, self-managed blocking is perceived more effective than externally imposed restrictions—an insight relevant to educators, organizations, and developers seeking to mitigate the costs of digital distraction.

Limitations and Future Plan

As part of a larger project, this study offers opportunities for future research. *First*, the use of an English-language reading task may limit generalizability to other task types and linguistic contexts. Future research can examine diverse task domains and native-language contexts to validate the robustness of identified attentional mechanisms and outcomes. *Second*, while eye-tracking provided fine-grained attentional measures, future studies can incorporate additional physiological and behavioral measures to further examine the cognitive mechanisms underlying digital distraction and attentional regulation. *Last*, although the sample size of our study is consistent with prior eye-tracking research, a larger sample can improve statistical power for predicting downstream behavioral outcomes. Future research could replicate the findings with larger and more diverse samples.

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